

Telecom Customer Segmentation Data Analysis Report

Exploratory Data Analysis, Feature Engineering, and Preprocessing

Executive Summary

This report outlines the initial phases of a customer segmentation project to understand customer characteristics and identify potential predictors of churn. Through comprehensive exploratory data analysis (EDA) and feature engineering, we have prepared a dataset for subsequent K-cluster analysis. Key findings highlight the nonlinear effects of tenure and monthly charges on customer behavior, informing the creation of actionable customer groups. The data has been cleaned, preprocessed, and split into training and testing sets, providing a solid foundation for building our K-means model and customer segment profiles.

Initial Data Assessment

The original dataset comprises 7,043 rows and 10 columns, representing a substantial number of customer records and relevant attributes. A thorough check for missing values revealed no null entries across any of the features. This indicates a clean dataset, eliminating the need for imputation strategies at this stage. The dataset has a mix of data types suitable for customer analysis:

Table 1
Dataset Summary

Variable	Data Type	Description
gender	Categorical	Gender of the customer (male or female)
Dependents	Categorical	If the customer has any dependents (yes/No)
PhoneService	Categorical	Customer has phone service or not (Yes/No)
MultipleLines	Categorical	Customer has multiple lines with the telco (Yes/No)
InternetService	Categorical	Internet service type subscribed (DSL or Fiber optic)
Contract	Categorical	Contract type (Month-to-month, One year or Two year)
Churn	Categorical	Customer churned or retained (Yes/No)
SeniorCitizen	Numerical	Customer is a senior citizen or not (1=senior citizen, 0 otherwise)
tenure	Numerical	How long the customer has been in this business
MonthlyCharges	Numerical	Amount a customer pays per month
Source: Canva		

Exploratory Data Analysis (EDA)

Univariate Analysis

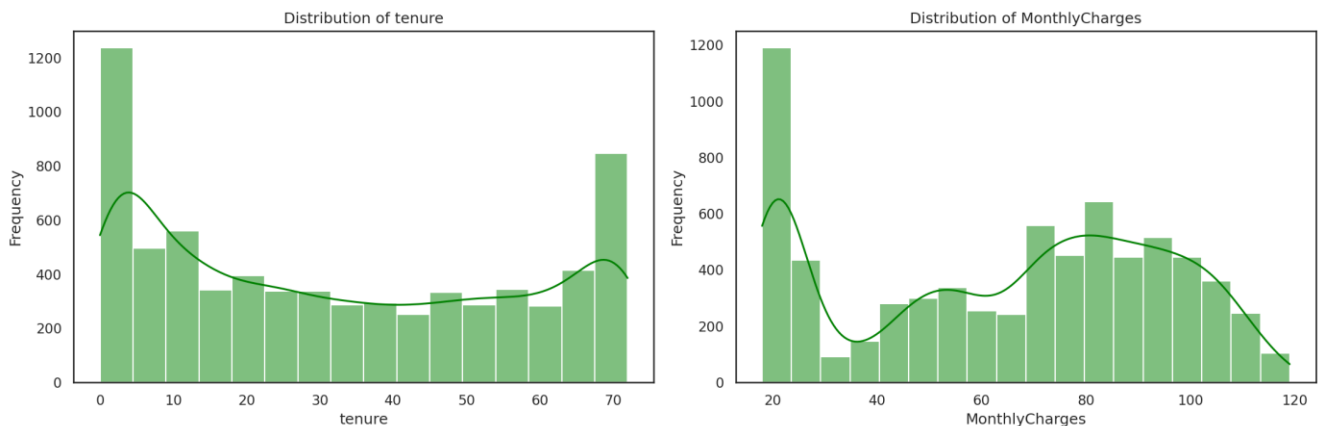
The histogram for tenure exhibits a multimodal distribution, with noticeable peaks at both lower, around 1-5 months, and higher tenure values, 60-72 months. This suggests distinct customer lifecycle stages, new customers and long-term loyal customers, rather than a single, uniformly distributed population.

Similarly, the MonthlyCharges histogram shows a multimodal distribution, with peaks in the lower-charge ranges (basic users) and the higher-charge ranges (premium users). This indicates varied service consumption patterns and price points across the customer base.

The presence of multimodal distributions for tenure and MonthlyCharges is a critical finding. It strongly suggests that treating these variables as purely continuous might obscure underlying customer segments and their unique behaviors. This insight is foundational for our decision to create grouped versions of these features, as discussed later.

Figure 1

Distribution Plots



Bivariate Analysis

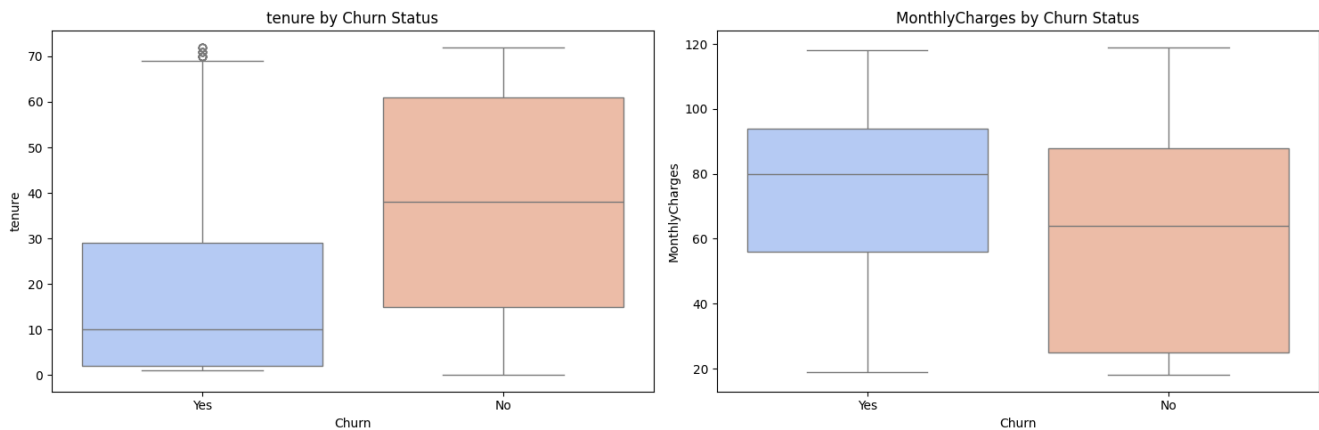
Numerical Features vs. Churn

The box plots visually suggest differences in MonthlyCharges between churned and retained customers. The independent samples t-test confirms a statistically significant difference (T-statistic: 18.39, P-value: 0.000). This implies that customers with higher monthly charges are more likely to churn.

Similarly, box plots and the t-test (T-statistic: -34.82, P-value: 0.000) demonstrate a statistically significant difference in tenure between churned and retained customers. This indicates that customers with lower tenure are more prone to churning or have already churned.

Figure 2

Numerical Features vs Churn Plots

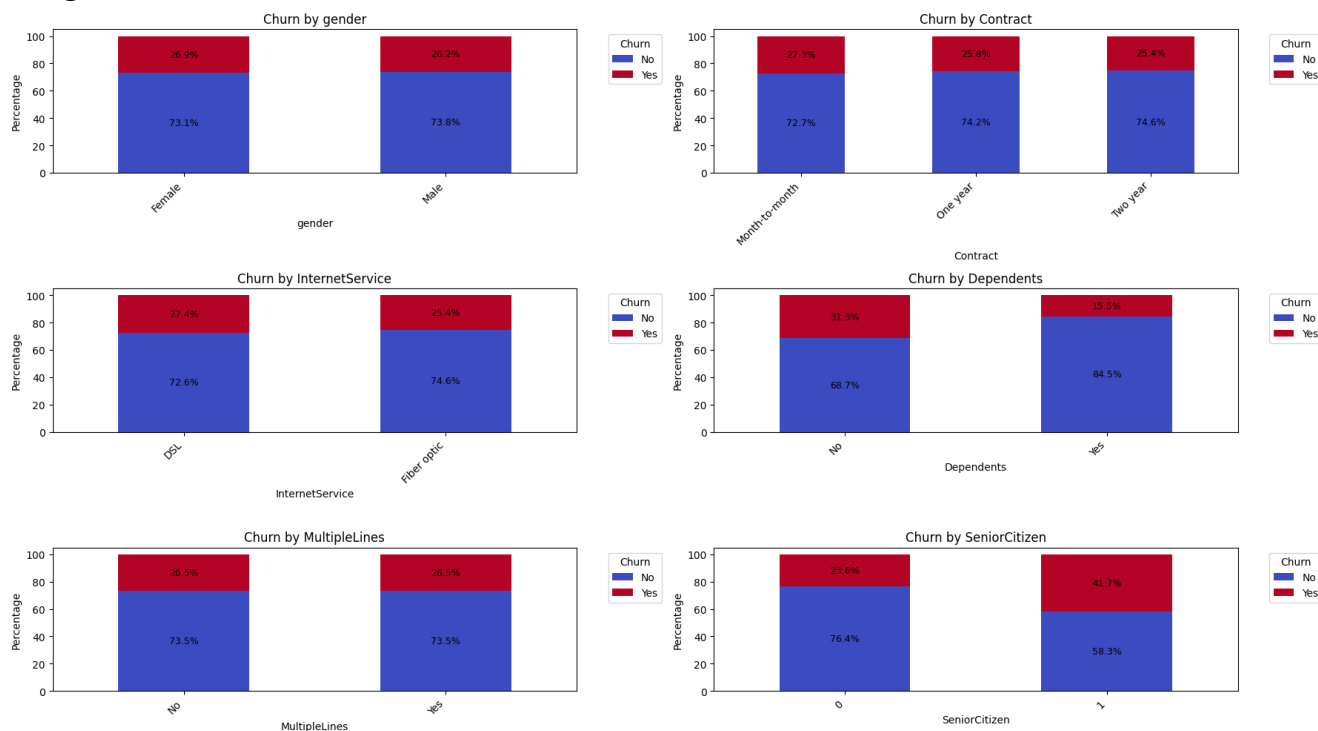


Categorical Features vs. Churn

The Chi-Square test reveals a statistically significant association (P-value: $4.92e-43$). Customers with no dependents show a higher churn rate compared to those with dependents. While visual inspection might suggest differences, the Chi-Square test indicates no statistically significant association (P-value: 0.241). This implies that the type of contract (Month-to-month, One year, Two year) is not independently associated with churn at a statistically significant level within our customer base, contrary to common business assumptions, which could be an interesting finding for further investigation.

The Chi-Square test shows no statistically significant association (P-value: 0.064). This suggests that the type of internet service (DSL, fiber-optic) itself is not a primary independent driver of churn. The Chi-Square tests for gender and MultipleLines and PhoneService indicate no statistically significant association with churn. This suggests these features are not strong predictors for churn in this dataset.

Figure 3
Categorical Variables vs Churn



Feature Engineering

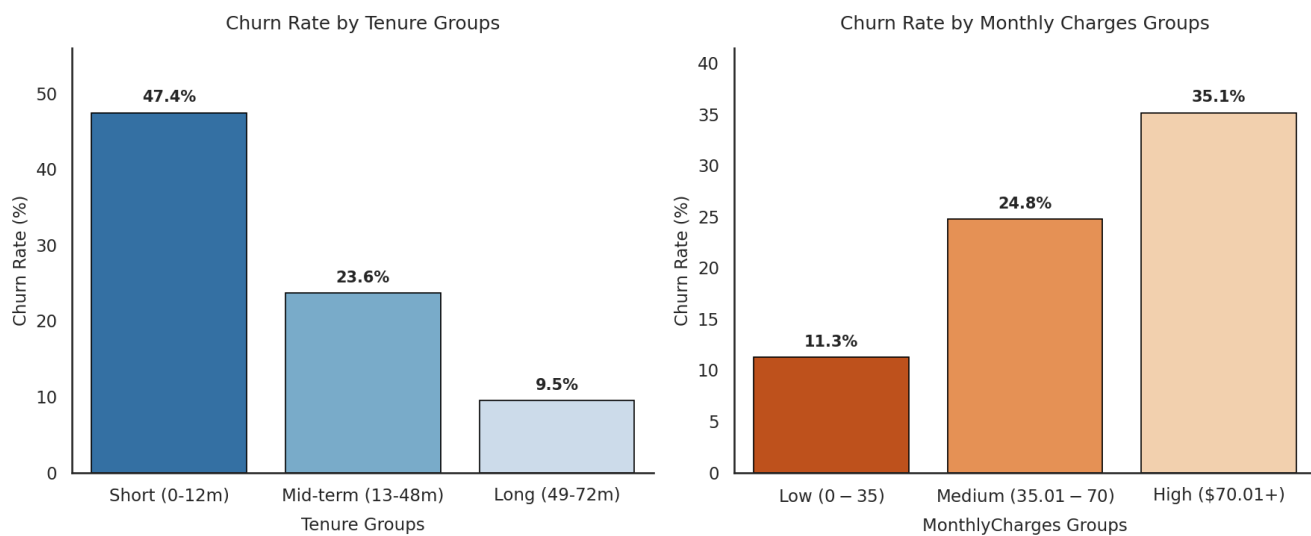
Building on the insights from the EDA, particularly the multimodal distributions of tenure and MonthlyCharges, we created derived categorical variables to improve interpretability and actionability for business intelligence.

Raw numerical ranges, while precise, are often less intuitive for business stakeholders. By creating groups like Short (0-12m), Mid-term (13-48m), Long (49-72m) for tenure, and Low (\$0-\$35), Medium (\$35.01-\$70), High (\$70.01+) for MonthlyCharges, we converted granular data into easily understandable groups. These segments directly inform targeted marketing, sales, and customer service strategies. For instance, a Short Tenure customer requires different onboarding and retention efforts compared to a Long Tenure loyal customer.

Moreover, the observed multimodal distributions for both tenure and MonthlyCharges indicate that the customer base is not monolithic. There are naturally occurring clusters of customers at different points along these scales. Our grouping strategy acknowledges these inherent divisions, ensuring that our segments reflect these real-world customer archetypes. For example, the distinct peaks in tenure directly inform the short, mid-term, and long tenure groups, which often correspond to different phases of the customer journey (acquisition, growth, loyalty).

Applying the Chi-Square test to these newly created groups against Churn confirmed a strong association. The tests showed that customers in their first year are at significantly higher risk of churn. Additionally, customers with higher monthly charges have a higher churn rate, suggesting potential issues with perceived value or competition at higher price points.

Figure 4
Derived Groups vs Churn Rate



Feature Selection and Data Preprocessing

Based on the detailed EDA, statistical tests, and machine learning fundamentals, the following features were selected for the K-Means model: Dependents, Tenure_Groups, and MonthlyCharges_Groups.

These features showed a statistically significant association with Churn in the bivariate analysis. The grouped features were selected over their continuous counterparts to simplify interpretation of customer segments and to avoid multicollinearity, given the strong correlation between continuous variables and their binned versions. Variables such as gender, Contract, InternetService, PhoneService, and MultipleLines were excluded because their association with Churn was not statistically significant in our tests, thereby simplifying the clustering model and focusing on the most impactful features.

One-Hot Encoding

Because the K-Means clustering algorithm typically requires numerical input, the selected categorical features (Dependents, Tenure_Groups, MonthlyCharges_Groups) were transformed using One-Hot Encoding. This process converts each categorical value into a new binary (0 or 1) column, ensuring the algorithm can properly process these attributes. The resulting encoded dataframe has 8 features. Notably, the Churn variable was excluded from the data at this stage because our unsupervised model does not require a target variable.

Data Splitting for K-Means Clustering

For unsupervised learning algorithms like K-Means, which do not rely on a target variable (y), the feature set X was split directly into training and testing sets. This enabled model development on one subset and evaluation of cluster stability and performance on the unseen subset.

Training Set (X_train): Comprises 5,634 rows and 8 columns (80% of the data).

Testing Set (X_test): Comprises 1,409 rows and 8 columns (20% of the data).

Conclusion

This analysis established a strong foundation for customer segmentation by identifying meaningful patterns in our customer base. The EDA and statistical tests showed that tenure, MonthlyCharges, and Dependents have significant relationships with churn. In contrast, gender, contract type, internet service, phone service, and multiple lines did not show statistically significant associations in this dataset. The multimodal distributions of tenure and MonthlyCharges also revealed distinct customer behavior and spending patterns. Based on these findings, the selected features were one-hot encoded and prepared for K-Means clustering, resulting in an eight-feature dataset that was split into training and testing sets.